

Annotation guidelines

Classification Criteria: This subsection outlines the categorization of the patient's psychosocial environment.

Label: COMMUNITY-PRESENT

Assessed as **True** if the narrative indicates positive social integration, robust support systems, or any kind of (non-negative) familiar relationship, including pets (e.g., kinship, marriage, friendships).

- *Reside con su cónyuge.* ('They reside with their spouse.')
- *Mantiene una relación cercana con sus hijos de 27 y 25 años.* ('They maintain a close relationship with their children, ages 27 and 25.')

False otherwise.

Label: COMMUNITY-ABSENT

Assessed as **True** should the text denote social isolation, bereavement, or deleterious interpersonal circumstances such as divorce, abuse, or the passing of a family member.

- *Casado pero legalmente separado; reside con compañeros de piso.* ('Married but legally separated; he resides with roommates.')
- *Se encuentra en estado de viudez.* ('They are widowed.')

False otherwise.

Label: EDUCATION

Identified as **True** if there is explicit reference to the patient's schooling, training, or attained diplomas or degrees.

- *Sufrió una lesión tras haber completado sus estudios secundarios.* ('They suffered an injury after completing their secondary education.')
- *Posee un doctorado en ingeniería industrial.* ('They hold a doctorate in industrial engineering.')

False otherwise.

Label: JOB STATUS

Categorized according to the patient's **current** professional activity:

- **True:** Currently engaged in employment.
- **False:** Currently inactive in the workforce (e.g., unemployment, retirement, housewife (*ama de casa*), or disability).
- *Empleada del hogar.* ('She works as a domestic employee.')
- *Sirvió en el ejército durante 15 años, actualmente jubilado.* ('He served in the military for 15 years and is currently retired.')

A PRESENT VERBAL TENSE should be taken into account for identifying the job status as True or False, unless it's mentioned in the end of the note (the discharge, *el alta*).

Unless explicitly mentioned, do not classify as any of these (for instance, even if a person is stated as a student, they may or may not have some kind of Job; on another note, if past

job is mentioned but current is not mentioned, we do not have current information either). In these cases or others in which no mention is made, tag as **None**. However, if the patient is said to return to work after the discharge from the hospital, the note is tagged as **True**.

Even if the patient is said to have Job Status **True**, if they deacease at the end, tag as **None** (information no longer relevant).

Label: HOUSING

Distinguishes between stable domesticity and precarious living conditions:

- **True:** Permanent or stable residency.
- **False:** Homelessness, transient living, or inadequate housing.
- *Reside de forma permanente en una estancia para la tercera edad.* ('They permanently reside in a senior living facility.')
- *En situación de calle, a veces duerme en la casa de un amigo.* ('They are homeless; they sometimes sleep at a friend's house.')

If one of the cases appears NEGATED, the other can be assumed (based on a present temporal tense). If the patient is said to return to their home after discharge from the hospital, the note is tagged as **True**.

Even if the patient is said to have Housing True, if they deacease at the end, tag as None (information no longer relevant). Otherwise, the tagging should be done **INDEPENDENT OF THE VERBAL TENSE**, taking into account the **LAST KNOWN INFORMATION**.

None otherwise.

Label: ALCOHOL, TOBACCO, AND SUBSTANCE USE

Assesses the patient's history with controlled substances or alcohol:

- **Present:** Active consumption at the time of reporting. Must have some kind of indication of actuality, like present verbal tense or adverbs like *Currently*.
- **Past:** Historical consumption followed by cessation. The verbal tense of the phrase should be taken into account to avoid abstract interpretations.
- **None:** Explicit denial of use or total absence of mention.
- *Cesación tabáquica hace 15 años.* ('They quit smoking 15 years ago.')
- *Uso continuado de opiáceos.* ('Continued opioid use.')

Even if the patient is said to have ALCOHOL/TOBACCO/SUBSTANCE as **Present**, if they deacease at the end, tag as **Past** (information no longer relevant but still tagged as **Past**).

Named Entity Recognition (NER): The following entities should be tagged within the text to identify specific psychosocial indicators, even if they pertain to a person other than the patient, but only if they relate to SDOH factors. It should be minimized as much as possible, as we are interested in specific

keywords rather than long tags, but some tags may need extra information to be considered of a specific tag, or the extra words may be of importance. Example: ABUSO can be of different types, so tag only WITH extra words: *abuso físico* ('physical abuse'), for COMMUNITY-ABSENT, or *abuso de tabaco* ('tobacco abuse'), for TOBACCO (only tag 'tabaco' in this case). Moreover, do not tag words that may be too general; e.g. *trauma* ('trauma'), *peligro* ('danger'), *compañero* ('companion'), *relación* ('relationship')... On the other hand, though most keywords will be nouns or verbs, some may appear in adjectives such as *divorciado* ('divorced') or *escolar* ('school').

- **COMMUNITY-PRESENT:** Identify all mentions of family, partners, and social allies, including pregnancies and pets. If a relationship is explicitly denied (e.g., *sin hijos* ('without children')), apply the **NEGATION** modifier.
- **COMMUNITY-ABSENT:** Identify mentions of separation (including breakups), divorce, abuse, or bereavement (e.g., *fallecido* ('deceased')), including abortions. In this specific topic, more abstract expressions like *se siente solo* ('they feel alone') should be annotated.
- **EDUCATION:** Extract any reference to educational institutions or levels of attainment, including classmates and teachers. When talking about institutions, only words explicitly related to education levels have to be tagged, not general statements about qualifications, institution category (*private*, *public*...) or academic performance.
- **JOB STATUS:** Tag any indication of employment or lack thereof, including professions, ownership of businesses, or generic statements like *trabajo* ('work'). For former roles, classify also, but some kind of **TEMPORALITY** or **NEGATION** may have to be applied.
- **HOUSING:** Tag all references to domestic environments (*hogar* ('home')), shelters, or the absence of a fixed address. **NEGATION** may be applicable in cases like *No posee vivienda fija* ('Does not have a stable place to live').
- **SUBSTANCE USE:** Identify mentions of specific substances and habits (e.g., *exfumador* ('former smoker')). Note: In the case of DRUGS, this refers to illicit drugs or clinical dependency, not standard medications, even if a dependency is mentioned.

Supplementary Tagging (Modifiers): These tags must only be applied in conjunction with the primary entities listed above:

- **NEGATION:** Applied when an entity is mentioned only to be denied. Tag it only in words already having some kind of SDOH NER tag, not by itself or with **TEMPORALITY** or **QUANTITY**; we want to negate an explicit expression, but we are not interested in the actual words expressing negation. WARNING: *exfumador* ('former smoker') is NOT negated, but *dejar de fumar* ('to stop smoking') is!. Negation always has to have some kind of negation trigger or cue:
 - Adverbs and determiners: *no* ('no/not'), *ni* ('nor'), *nunca* ('never'), *jamás* ('never (ever)'), *nada* ('noth-

ing'), *nadie* ('nobody'), *ninguno/a* ('none/neither'), *tampoco* ('neither/either')

- Prepositional: *sin* ('without'), *excepto* ('except'), *salvo* ('except'), *menos* ('except')
- Verbal triggers indicating cessation or refusal: *dejar de* ('to stop'), *parar de* ('to stop'), *cesar de* ('to cease'), *negar* ('to deny'), *rechazar* ('to reject')

- **TEMPORALITY:** Encompasses all temporal markers, both numerical and abstract ones (e.g., *hace 5 años* ('five years ago'), *una vez por semana* ('once a week'), *historial de (...)* ('history of (...)'). Tag as much as possible, but DO NOT tag words that do not necessarily mean anything temporal, like *intermitentemente* ('intermittently').
- **QUANTITY:** Pertains to volume of intake or numerical demographics (e.g., *dos paquetes diarios* ('two packs a day'), *tres hijos* ('three children')). Do NOT tag abstract words, like adverbs, only explicit numerical quantities, and include prepositions and similar structures that also contribute to the information, like *desde hace cinco años* ('for the past five years').

In cases like *5 cigarrillos/día* ('5 cigarettes/day'), where **QUANTITY** and **TEMPORALITY** may be mixed, tag it as the first tag that appears (in this case, quantity).